



**UNIT MATEMATIK
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KEMENTERIAN PENDIDIKAN MALAYSIA**

SECTION A

1. Evaluate the limits of the following, if it exists:

a) $\lim_{x \rightarrow 1^-} \frac{x^2 + x - 2}{|x - 1|}$ [4 marks]

b) $\lim_{x \rightarrow +\infty} \sqrt{x}(\sqrt{x+1} - \sqrt{x})$ [5 marks]

2. a) If $y = \sin(x^2 + 1)$, show that $x \frac{d^2 y}{dx^2} - \frac{dy}{dx} + 4x^3 y = 0$ [5 marks]

- b) Find $\frac{dy}{dx}$ for each of the following

i) $y = (\ln x)^6$ [2 marks]

ii) $y = e^{3x}(2x^4 + 1)$ [3 marks]

3. Given $f(x) = x^3 - 3x^2 - 9x + 11$. Find the critical point of $f(x)$ and determine whether the critical points are local minimum or maximum. [6 marks]

SECTION B

1. Given two complex numbers $z_1 = 2 + i$ and $z_2 = 1 - 2i$.

a) Determine the value of k if $\frac{1}{z_1} = k \overline{z_1}$ [4 Marks]

b) Express $\frac{z_1^2}{z_1 - z_2}$ in the form of $a + bi$ where a and b are real numbers [5 Marks]

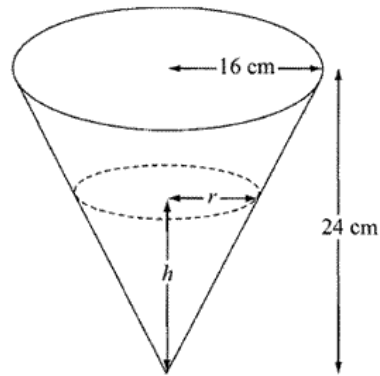
2. Solve the following equations

a) $\log(x + 2) + \log(x - 1) = 1$ [6 Marks]

b) $|2 - x^2| < 2x - 1$ [7 Marks]

3. Given that $g(x) = \frac{1}{x}$ where $x \neq 0$ and $h(x) = \sqrt{2-3x}$,
- a) State the domain and range of $h(x)$. [2 marks]
- b) Find $g \circ h(x)$ and $g \circ h^2(x)$. Hence, evaluate $g \circ h^2(0)$ [6 marks]
4. Given $f(x) = \ln(2x-1)$
- a) If $h(x) = f(2x+3)$, find $h(x)$ [3 marks]
- b) Find $h^{-1}(x)$, state the domain and range of $h^{-1}(x)$. [4 marks]
- c) Sketch the graph of $h(x)$ and $h^{-1}(x)$ on the same diagram [3 marks]
5. a) Given $f(x) = \begin{cases} \frac{x^2-1}{1-x} & \text{if } x > 1 \\ -2 & \text{if } x \leq 1 \end{cases}$
- Determine the continuity of f at $x = 1$. [5 marks]
- b) Given $f(x) = \begin{cases} Ax^2 + 2, & x \leq 3 \\ Bx + 1, & 3 < x < 5 \\ 4, & x \geq 5 \end{cases}$
- is continuous over $[-1, 6]$ with A, B constants. Evaluate the values of A and B . [6 marks]
6. a) Given $x = \frac{t}{1-t}$ and $y = 1+t$,
- i) Find $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$ in terms of t . [7 marks]
- ii) Find the value of $\frac{dy}{dx}$ when $x = -2$. [3 marks]
- b) Given the equation $x^2 + 4xy + y^2 + 27 = 0$. Find the value of $\frac{dy}{dx}$ when $x = -3$. [6 marks]

7.



A container is made in the shape of a hollow inverted right circular cone. The height of the container is 24 cm and the radius is 16 cm as shown in figure. Water is flowing into container. When the height of water is h cm, the surface of the water has radius r cm and the volume of water is $V \text{ cm}^3$.

a) Show that $V = \frac{4\pi h^3}{27}$. [4 marks]

b) If the water flows into the container at a rate of $8 \text{ cm}^3 \text{ s}^{-1}$. Find in terms of π , the rate of h which $h = 12$. [4 marks]

END OF QUESTION PAPER